

# The Probe Is the Bottleneck: Zero-Shot Round-Trip Recovery

## Does Not Yet Certify Emotion Controllability in Text-to-Image Generation

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### Abstract

Round-trip recovery is the default certificate of emotion controllability in text-to-image generation: condition on an emotion, classify the generated image, count the matches. Zero-shot CLIP makes the loop free, so the loop is everywhere. We test whether it certifies anything. Our protocol holds image content independent of the intended emotion and bounds every number with a chance floor, a template ceiling, a shuffled-label permutation control, family-wise correction, and probe error deconvolved into the estimate. A CPU pilot (SD-Turbo, CLIP ViT-B/32,  $n = 14$  per tier) returns a guarded null: no conditioning tier beats the shuffled-label control (Holm-adjusted  $p = 0.27$ ), and the emotion tier’s apparent lift, 0.214 over a 0.143 chance floor, corrects to 0.165 once the probe’s measured error is removed. The failure is the instrument. In domain the probe collapses onto neutral, using 2 of 7 labels and assigning neutral to 90% of items; on real scenes it reads emotion at only 40.3% (ViT-L/14: 45.5%, McNemar  $p = 0.038$ ). We release AffectBench, the harness, and a preregistered powered run ( $n = 420$  per tier,  $\geq 95\%$  power) that executes only after a probe demonstrates in-domain competence.

## 1 Motivation

Emotion-conditioned image generation now has a name and a leaderboard. EmoGen defined the task and the metric the field defaults to: generate on an emotion, classify the image, count the matches (Yang et al., 2024). EmotiCrafter extended conditioning to continuous valence and arousal (Dang et al., 2025); adapters and guidance followed (Yuan et al., 2025; Xia et al., 2025); training-free prompt conditioning arrived with EPIG (Othmen et al., 2026). The applications are not hypothetical: affective media, accessibility, tools that respond to how a sentence feels rather than what it names.

Every such system faces one question. Did the image convey the emotion, or did we only intend it to? Humans answer slowly and expensively. The round trip answers instantly, and with zero-shot CLIP it answers for one forward pass per image. That cheapness is exactly why the assumption deserves a hard test: a free measurement spreads before anyone checks what it measures. Generative evaluation has been here before. FID contradicts human raters on modern text-to-image models (Jayasumana et al., 2024). Most fidelity metrics fail basic sanity checks (Naeem et al., 2020). No standard metric strongly tracks human judgment at scale (Stein et al., 2023). Emotion round-trip recovery has never received the same scrutiny. This paper supplies it, in runnable form.

## 2 Gap

Three omissions make published round-trip numbers unreadable.

No controls. Accuracy is reported bare: no chance floor, no template ceiling, no test against random label agreement. The failure mode is obvious once stated. If conditioning paints a fixed per-emotion scene and the probe is worded to match it, recovery is a tautology; the number measures prompt-probe vocabulary overlap, not the image.

No instrument validation. The classifier doing the recovering is trusted, not tested. Zero-shot CLIP gets dropped in because it is free, despite image-emotion performance that is weak on abstract art (Widhoelzl et al., 2024) and low enough on standard benchmarks that recent work names the visual affective gap as the problem itself (Wu et al., 2025). No one prints the probe’s in-domain error beside the recovery number it produced, and no one propagates that error into the estimate.

No preregistration. Runs are sized and analyzed after the fact. A null becomes indistinguishable from an underpowered shrug; a positive, from a garden of forking paths.

The metric-critique tradition closed these gaps for FID and Inception Score. For emotion controllability they stand open, and they matter more here: perceived emotion is subtler than fidelity, and the default probe is weaker.

### 3 Hypothesis

The hypotheses are frozen in the repository’s preregistration before the powered run. The committed pilot is a calibration of the instrument, not a test of them.

- H1 (conditioning beats chance). For each conditioning tier (naive, emotion, affect), recovery accuracy exceeds the shuffled-label permutation null (one-sided, 2000 permutations).
- H2 (conditioning beats the template). The emotion and affect tiers add recovery over the scene template ceiling on the content track.
- H0 (the registered null). Neither holds after multiple-comparison control.

One outcome-neutral condition governs all three, in the Registered Reports sense (Chambers, 2013): the probe must demonstrate in-domain competence before any hypothesis test is interpretable. A probe collapsed onto two of seven labels scores chance on balanced data whatever the images show, so a collapsed run is reported as instrument failure, never as evidence for H0.

### 4 Research questions

- RQ1. Does prompt-level conditioning raise recovered-emotion accuracy above the shuffled-label null, per tier, under Holm correction?
- RQ2. Does any candidate probe clear an in-domain competence gate: accuracy materially above the majority baseline on real emotional scenes, more than two of seven labels used on generated ones? Candidates: CLIP ViT-B/32, ViT-L/14, and an EmoSet-trained classifier through the same interface.
- RQ3. Does any apparent lift survive Rogan-Gladen correction for the probe’s measured error?
- RQ4 (secondary). On the text track, where valence and arousal are grounded in the sentence rather than a per-emotion prior, do continuous affect correlations show signal that categorical recovery misses?

### 5 Methodology

#### 5.1 Pipeline and conditioning tiers

Text enters a DistilRoBERTa emotion classifier (Hartmann, 2022) over the six Ekman emotions plus neutral (Ekman, 1992). The emotion is grounded in Russell’s circumplex (Russell, 1980): valence and arousal come from a per-emotion prior blended with lexical affect from

| Tier    | Prompt composition                           | Role                               |
|---------|----------------------------------------------|------------------------------------|
| raw     | content + style                              | negative control, chance floor     |
| naive   | content + bare emotion word                  | minimal baseline to beat           |
| emotion | content + engineered mood phrase             | the method, categorical            |
| affect  | emotion tier + palette, lighting from $v, a$ | the method, dimensional            |
| scene   | fixed per-emotion template, no content       | positive control, template ceiling |

Table 1: Conditioning tiers. Content is independent of emotion by construction.

an affect lexicon (empirical norms: Warriner et al., 2013; Mohammad, 2018; the shipped default is a small offline demo) as  $v = c v_{\text{lex}} + (1 - c) v_{\text{prior}}$ , with lexical coverage  $c$  capped at 0.8 so a fully in-lexicon input never discards the prior. Scoring skips function words and applies a two-token negation flip to valence. The blend is a stated heuristic, ablatable at  $c = 0$  and  $c = 1$ , and flagged as an open validity question.

Generation is SD-Turbo (Sauer et al., 2023) on a frozen backbone. Conditioning is prompt-level only and increases over the tiers in Table 1, which are the ablation. Content is never selected by the emotion; the same subject renders under every emotion, and seeds are shared across tiers per item, so contrasts are paired on generation noise.

## 5.2 Recovery and the circularity attack

A swappable probe reads emotion and graded affect back from the image. The default is CLIP zero-shot (Radford et al., 2021) with two fixes: emotion logits scaled by the model’s learned temperature, and valence and arousal taken as expectations over an ordered ladder of anchor prompts at a milder fixed temperature. Probe templates are frozen before measurement.

Circularity, the risk that a CLIP-family probe simply agrees with the emotion word the conditioning wrote, is attacked three ways. Decoupled content forces any signal through the mood modifier, not the depicted scene. The shuffled-label permutation control quantifies the circularity baseline directly: recovery counts only when it clears random target reassignment. And an independent non-CLIP image-emotion probe slots into the same interface, removing the shared vocabulary entirely; validating the headline under it is a prerequisite, not an option.

## 5.3 AffectBench

The content track uses twenty affect-neutral subjects. The text track uses AffectBench, built on demand from GoEmotions (Demszky et al., 2020) under the official Ekman grouping: balanced per-class sampling from the test split, exact within-sample dedup, subtraction of any item whose normalized text appears in the train split, counts and a content hash logged, composition and licensing documented in a datasheet (Geburu et al., 2021). GoEmotions is Reddit-English with modest Ekman-level agreement ( $\kappa \approx 0.33$  to 0.44); the benchmark inherits that skew, and the scope is text-prompt controllability, stated as such.

## 5.4 Statistics

Every tier reports recovery accuracy with a bootstrap 95% CI, macro-F1, valence and arousal correlations with CIs, CLIP-T for content preservation, the majority-class baseline, a probe-collapse diagnostic (how many labels the probe ever uses), and the shuffled-label  $p$ -value. Holm correction controls the family-wise rate across tiers (Holm, 1979); Cohen’s  $h$  accompanies every contrast, so significance never travels without magnitude. Because the probe is imperfect, apparent recovery is deconvolved through the Rogan-Gladen estimator (Rogan and Gladen, 1978) using the probe’s measured sensitivity and specificity: instrument error becomes a correction, not a caveat. Provenance is total. Each run logs git SHA, library versions, pinned model revisions, device, dtype, and the benchmark hash.

| Tier    | Accuracy [95% CI]    | Macro-F1 | Shuffled $p$ | Holm $p$ |
|---------|----------------------|----------|--------------|----------|
| raw     | 0.143 [0.000, 0.357] | 0.038    | 0.86         | n/a      |
| emotion | 0.214 [0.000, 0.429] | 0.112    | 0.23         | 0.27     |
| affect  | 0.214 [0.000, 0.429] | 0.133    | 0.14         | 0.27     |
| scene   | 0.286 [0.000, 0.575] | 0.184    | 0.14         | n/a      |

Table 2: Pilot recovery by tier. Chance and the majority baseline coincide at 0.143 (1/7).

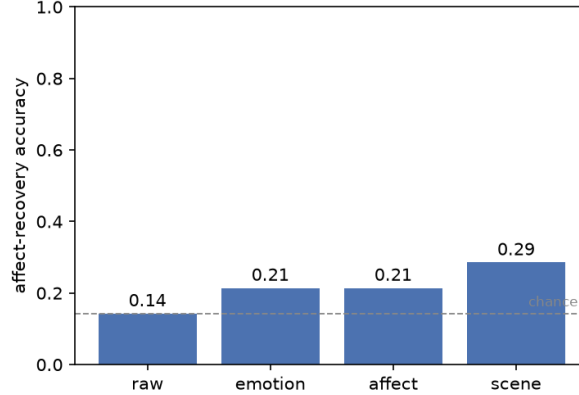


Figure 1: Recovery accuracy by tier in the committed pilot. Every interval includes the 1/7 chance line.

### 5.5 The pilot as proof of concept

The repository ships a committed CPU pilot (256 px, 2 subjects, 1 seed;  $n = 14$  per tier,  $n = 7$  for scene; the naive tier postdates the committed pilot and has no committed numbers yet). Its purpose is calibration: run the protocol end to end and measure the instrument. It did both, and the second finding governs everything.

Read four things together in Table 2 and Figure 1. First, the probe collapsed in domain: 2 of 7 labels used, neutral assigned to 90% of generated items (Figure 2). A collapsed probe scores chance on raw by construction, so the floors are necessary but not sufficient; the collapse diagnostic and majority baseline print beside every number for exactly this reason. Second, the lift is one image. Emotion over raw is +0.071 (Cohen’s  $h = 0.187$ ,  $p = 0.255$ ), and affect adds nothing over emotion, as expected on the content track, where affect cues are a deterministic function of the emotion prior. Third, no tier clears the shuffled-label control. Fourth, correction sharpens the verdict rather than softening it: the emotion tier’s apparent 0.214 becomes 0.165 under Rogan-Gladen, indistinguishable from chance at this  $n$ , attributing most of the apparent lift to the instrument’s error structure.

Probe validation locates the failure. Out of domain (faces,  $n = 200$ ), ViT-B/32 reads Ekman emotion at 29.0%, with bimodal recall: neutral 0.81 and anger 0.61 against surprise 0.04, fear 0.06, sadness 0.08. In domain on real scenes (EmoSet,  $n = 400$ ) it reaches 40.3% while emitting all seven labels; ViT-L/14 reaches 45.5% (faces: 37.5%), significant per set by exact paired McNemar test ( $p = 0.038$  on scenes, discordant 36 vs. 57;  $p = 0.040$  on faces). Two consequences follow. The probes are not blind on real photographs, so the neutral collapse is a property of the generated images (256 px, 2-step SD-Turbo) interacting with the probe; the generator’s affective weakness is a co-suspect. And a structural suspect sits underneath both: image-emotion resources standardize on Mikels’ eight categories with no neutral (Mikels et al., 2005; Yang et al., 2023), while the targets here are Ekman plus neutral, and neutral is exactly where the probe collapses.

The pilot did what a Stage 1 calibration should: it proved the harness runs, and it caught its own instrument failing before that instrument could manufacture a conclusion.

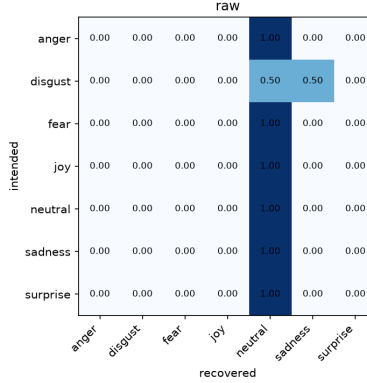


Figure 2: Raw-tier confusion from the pilot: predictions collapse onto the neutral row.

The repository makes the catch checkable, with pinned revisions, run manifests, a 200-test suite, a CI target that re-derives every reported number from the committed records, and a schema-validated submission format for scoring other generators and probes under the frozen protocol.

## 6 Timeline

The plan is staged as a Registered Report: methods and analysis frozen before the confirmatory data exist (Chambers, 2013; van Miltenburg et al., 2021). Table 3 gives the phases.

| Phase | Work                                                                                                                               | Gate to advance                                                                   | Status                                                               |
|-------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 0     | Build harness, floors, controls; run CPU pilot; commit artifacts                                                                   | Harness re-derives all numbers in CI                                              | Complete, committed                                                  |
| 1     | Gate candidate probes in domain: ViT-L/14, then an EmoSet-trained classifier via the probe interface                               | More than 2 of 7 labels on generated scenes; accuracy materially above majority   | In progress; L/14 registered candidate (45.5%, McNemar $p = 0.038$ ) |
| 2     | One GPU day: 20 subjects $\times$ 7 emotions $\times$ 3 seeds, $n = 420$ per tier (scene 21), 512px; frozen analysis of H1, H2, H0 | Stopping rule: one run at registered $n$ ; no optional stopping, no dropped tiers | Withheld until Phase 1 passes                                        |
| 3     | Human study (3+ raters, Cohen’s $\kappa$ vs. probe); cross-system runs (EmoGen, EmotiCrafter, CoEmoGen)                            | Rater recruitment; compute                                                        | Wired, unrun                                                         |

Table 3: Staged plan in Registered Reports form.

Power is fixed before running. At the ViT-L/14 in-domain ceiling of 0.455,  $n = 420$  per tier gives at least 95% power to detect even a weak conditioning effect (strength 0.2, recovery 0.205) at  $\alpha = 0.05$ ; a Phase 2 null is therefore a properly powered null. If the gated probe collapses in domain, the run is withheld and the probe replaced, not reinterpreted. Deviations are logged in the preregistration before the numbers are read. The result publishes whichever way it comes out.

## 7 Contributions

- A gated, preregistered round-trip protocol for emotion controllability: decoupled content, chance and template floors, a shuffled-label permutation control with Holm correction, effect sizes on every contrast, and probe error deconvolved into the estimate rather than footnoted beside it.
- AffectBench and an open harness: a frozen, datasheeted, dedup-checked GoEmotions-derived benchmark; pinned models, total run provenance, a 200-test suite with CI-locked numbers, and a submission schema for scoring any generator or probe under identical rules.
- A diagnosed negative result: zero-shot CLIP round-trip recovery, as currently practiced, cannot certify emotion controllability. The mechanism is measured, not asserted: in-domain neutral collapse, a 40.3% ceiling on real scenes, and an apparent lift that correction returns to within noise of chance. The finding warns a subfield off a free metric before it hardens into a leaderboard.

## 8 Limitations

The instrument is the binding limitation, and it scopes the central claim: nothing here shows emotion conditioning fails, only that this cheap certificate cannot show it succeeds. A chance-level result is uninformative on its own, since a collapsed probe scores the majority baseline regardless of the image; the collapse diagnostic prints beside every number for that reason, and the null is guarded rather than triumphant. Mood modifiers and the CLIP probe share affective vocabulary; the floors bound the leakage and the non-CLIP probe removes it, but that probe’s validation is Phase 1 work, not done. On the content track, valence and arousal are per-emotion priors, so the affect-versus-emotion contrast tests palette and lighting only; text-grounded affect is testable only on the text track. The pilot is one generator, one probe, one seed, fourteen items per tier; its intervals span most of the possible range by design, and cross-system claims await Phase 3. Generated-image quality is a co-suspect: 2-step, 256 px SD-Turbo may carry little affective signal for any probe to read, a confound the 512 px powered run reduces but does not remove. AffectBench inherits GoEmotions’ Reddit-English domain and modest annotator agreement; disgust is scarce under the Ekman collapse; and the Ekman-plus-neutral taxonomy sits at odds with the Mikels-8 space standard evaluators are trained on. The registered remedy is evaluation on the native taxonomy with an EmoSet-trained probe, the mapping stated rather than hidden.

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